Audio Plugin - User Manual & Technical Documentation

# LLAMAJA GC2020B

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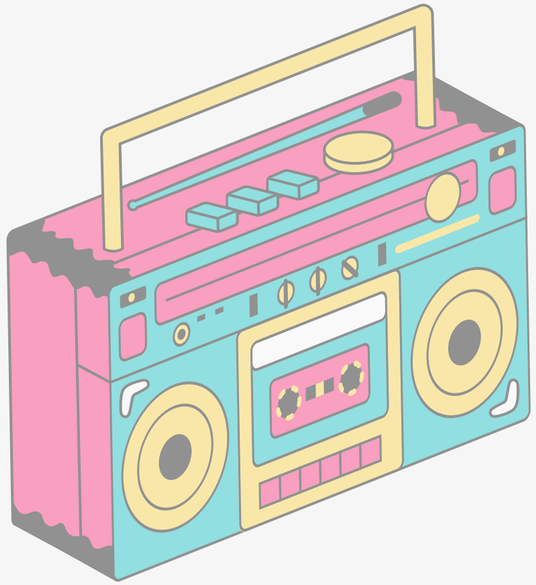
# INTRODUCTION



Welcome to Llamaja GC2020B II, a stereo compressor plugin inspired by classic dual-channel hardware dynamics processors. It combines flexible compression, gate control, stereo linking, and clear visual metering to provide musical and practical dynamic control for modern mixing workflows

**Why Llamaja GC2020B II?**

- **Dual-Channel Design: Independent CH1 and CH2 compression sections.**
- **Stereo Link: Linked behavior for stereo buses and mix processing.**
- **Built-In Gate: Expander/gate stage per channel for cleaner signals.**
- **Musical Dynamics Control: Threshold, ratio, attack, and release with hardware-style workflow.**

# FEATURES OVERVIEW

1

### 🎚️ Dual-Channel Stereo Compression

- Two fully independent compressor channels for CH1 and CH2.
- Ideal for dual mono processing or stereo material with precise channel control
- Link mode keeps both channels working together for stable stereo compression.

2

### 🚪 Integrated Expander / Gate

- Dedicated EXP GATE section on each channel for noise reduction and cleanup
- Helps control bleed, room noise, and low-level unwanted signal
- Useful for drums, guitars, vocals, bass, and dynamic live recordings

3

### 📊 Input Gain and Visual Metering

- Independent input gain control for proper signal feeding into the compressor.
- LED-style metering provides clear visual level feedback during operation.
- Makes threshold setup faster and helps maintain correct gain staging.

4

### 🛠️ Hardware-Inspired Dynamics Workflow

- Classic rack-style layout with direct access to Threshold, Ratio, Attack, and Release.
- Fast and intuitive control for both subtle leveling and stronger compression..
- Designed for practical use on tracks, buses, and stereo mix sources.

# FEATURES OVERVIEW

🎚️🎚️ **Dual-Channel Compression**

- Two independent compressor channels for left/right or dual-mono use.
- Stereo/dual mono operation for flexible routing.
- Ideal for tracks, buses, and stereo sources.

# FEATURES OVERVIEW

### 🚪 Integrated Expander / Gate

- Built-in expander/gate in each channel for noise reduction.
- Helps clean bleed, room noise, and low-level background signal.
- Useful on vocals, drums, bass, and guitars.

# FEATURES OVERVIEW

📉 **Core Compression Controls**

- Threshold sets when compression starts.
- Ratio defines how strongly the signal is reduced.
- Attack and release shape the response and recovery time.

# FEATURES OVERVIEW

📊 **Input, Output, and Metering**

- Input and output level controls support proper gain staging.
- Gain-reduction LED meter shows compressor activity clearly.
- Separate gate indicator helps monitor expander/gate action.

# FEATURES OVERVIEW

🔗 **Bypass and Link Control**

- Bypass lets you compare processed and dry signal quickly.
- Link mode keeps stereo compression behavior matched.
- Helps preserve stereo balance in mix-bus work.

# FEATURES OVERVIEW

### 🎛️ Musical Dynamics Shaping

- Smooth compression works well for transparent leveling.
- Can add punch, glue, or tighter control depending on settings.
- Suitable for both corrective and creative dynamics processing.

# INSTALLATION GUIDE

## STEP 1: DOWNLOAD

Download the latest .zip package (Mac or Win64) from the official release page or repository.

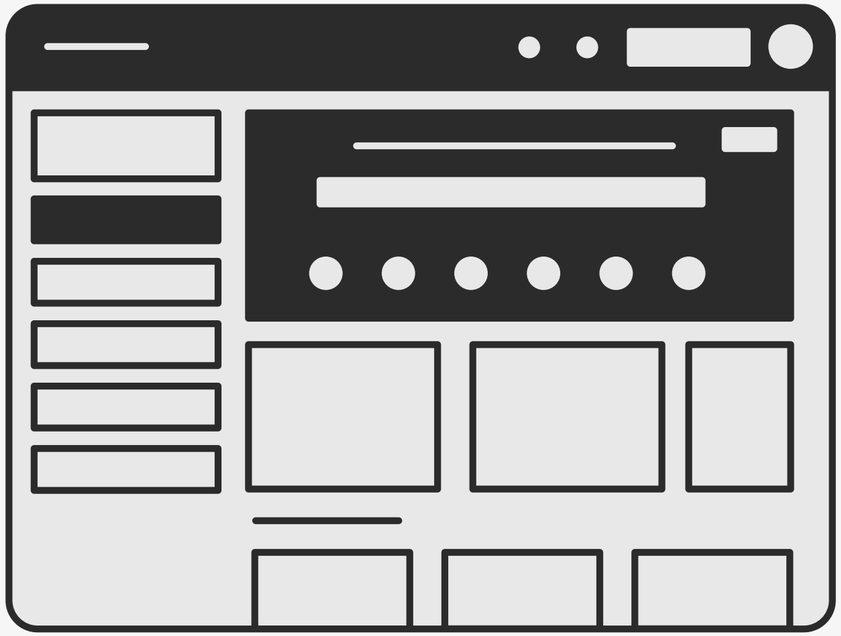
## STEP 2: INSTALL PLUGIN FILES

Move the plugin files to the appropriate directories.

- macOS: /Library/Audio/Plug-Ins/VST3 or /Components
- Windows: C:\Program Files\Common Files\VST3

## STEP 3: RESCAN PLUGINS

- Open your DAW
- Rescan plugin directory
- Locate "Llamaja GC2020B" in your plugin list



## STEP 4: ACTIVATE

Load the plugin on any audio track or bus

System Requirements

- Operating System: macOS 11.0 or later
- Architecture: Apple Silicon & Intel
- DAW Compatibility: All major hosts (Logic Pro, Fl Studio, Pro Tools, etc.)
- Plugin Formats: AU, VST3

# INTERFACE CONTROL

## Main Controls

### 🚪 EXP GATE
- Sets the gate / expander threshold per channel.
- Helps reduce low-level noise and background bleed.
- Useful for tightening drums, guitars, and vocals.

### 📉 COMP RATIO
- Defines how strongly the signal is compressed.
- Lower ratios sound smoother and more natural.
- Higher ratios give tighter dynamic control.

### ⏱️ RELEASE
- Sets how fast the compressor recovers.
- Short release adds movement and energy.
- Longer release sounds smoother and more even.

### 🔊 INPUT
- Adjusts the input level feeding the compressor.
- Important for proper gain staging and threshold response.
- Higher input can drive more compression.

### 🎯 THRESHOLD
- Sets the level where compression begins.
- Lower settings produce more gain reduction.
- Main control for overall compression amount.

### ⚡ ATTACK
- Controls how fast compression reacts to peaks.
- Fast attack clamps transients more aggressively.
- Slower attack preserves punch and impact.

### ✅ COMP IN
- Activates the compressor section for each channel.
- Useful for quick processed / unprocessed comparison.
- Helps check compression decisions while mixing.

### 📊 PK METER
- Displays gain reduction and peak activity visually.
- Helps monitor signal level and compression behavior.
- Useful for setting dynamics with fast visual feedback.

# INTERFACE CONTROL

## Main Controls

### 🔊 INPUT

• Adjusts the input level feeding the compressor.
• Important for proper gain staging and threshold response.
• Higher input can drive more compression.

### 🔗 LINK

• Links CH1 and CH2 for stereo operation.
• Keeps both channels responding together.
• Helps preserve stereo balance on buses and full mixes.

### 🟢 POWER

• Indicates that the plugin is active.[
• Confirms processing is available and ready to use.
• Matches the hardware-inspired visual workflow.

### 📊 PK METER

• Displays gain reduction and peak activity visually.
• Helps monitor signal level and compression behavior.
• Useful for setting dynamics with fast visual feedback.

### 🧩 CH1 / CH2

• Independent control sections for both channels.
• Can work separately or as a linked stereo pair.
• Useful for dual mono or stereo compression workflows.

# PRACTICAL APPLICATIONS

## VOCAL COMPRESSION 🎤

- Goal: Smooth dynamics and keep the lead upfront without pumping
- Settings: Ratio 2:1–4:1, Med-Fast Attack (3-10ms), Soft Knee
- Example: Light 2:1 on verses, pushing to 3:1 on choruses for consistency

## DRUM BUS COMPRESSION 🥁

- Goal: Glue the kit together and add punch while saving transients
- Settings: Ratio 1.5:1–3:1, Slow Attack (10-30ms), Synced Release
- Example: Try parallel compression (heavy 8:1 blended at 30-40%)

## GUITARS & BASS 🎸

- Goal: Even out levels and lock the low-end solidly in the mix
- Settings: Ratio 3:1–6:1, Med Attack (5-15ms), HPF on Sidechain (~80Hz)
- Example: Use medium attack to preserve pick noise, and tight release to follow the groove

# TECHNICAL SPECIFICATIONS

## AUDIO PROCESSING

- **Sample Rates:** 44.1kHz to 192kHz
- **Bit Depth:** 64-bit double-precision internal floating point processing.
- **Channels:** Stereo In / Stereo Out (True Stereo processing)
- **Anti-Aliasing:** Optional oversampling (up to 4x or 8x) to prevent distortion on fast release times.

## TECHNICAL FEEL

- **Transient response:** Extremely fast, capable of grabbing aggressive peaks.
- **Gain reduction action:** Smooth release curves that breathe musically with the tempo.
- **Control range:** Wide threshold and ratio options for maximum versatility.
- **CPU profile:** Highly optimized to allow dozens of instances in a single session without spiking CPU.

## PLUGIN SPECIFICATIONS

- **Framework**: JUCE 7/8
- **Code**: C++17
- **DSP Architecture**: Exact VCA gain-computer modeling and anti-aliased clipping stages.
- **Parameters**: AudioProcessorValueTreeState managed with smoothed values to prevent zippering during automation.

# TECHNICAL SPECIFICATIONS

## FX SPECIFICATIONS

- **Type:** VCA-style digital dynamics processor.
- **Core topology:** VCA-style digital dynamics processor.
- **Detection circuit:** RMS or Peak sensing (depending on your specific model implementation).
- **Knee type:** Hard knee for aggressive limiting and soft knee for transparent leveling.
- **Sidechain:** Internal filtering available to reduce low-frequency pumping.
- **Dynamics:** Smoothed parameter handling for artifact-free gain reduction.
- **Make-up gain:** Auto or manual gain compensation to match perceived loudness.
- **Stereo behavior:** Linked stereo operation to maintain a stable center image.
- **Bypass path:** True bypass-style plugin bypass at the host level.
- **Latency:** Zero latency design for real-time tracking and live mixing.

## AUDIO CHARACTERISTICS

- **Compression character:** Punchy, fast, and transparent at moderate settings.
- **Dynamic control:** Ranges from subtle vocal leveling to extreme drum bus smashing.
- **Coloration:** Clean and precise, retaining the transient impact of the source material.
- **Presence control:** Fast attack times effectively control harsh peaks and sibilance.
- **Mixing style:** Works perfectly on individual inserts, instrument busses, and the master fader.

# TROUBLESHOOTING

**Pumping / Unnatural Breathing**

- Increase the Release time to smooth out gain reduction.
- Reduce the Ratio or lower the Threshold for less aggressive compression.

**Loss of High-End / Dull Sound**

- Slow down the Attack time to let initial transients (stick hits, pick attack) pass through.
- Use the sidechain high-pass filter (if available) so the compressor doesn't overreact to bass frequencies.

**Plugin Not Appearing**

- Confirm correct installation path (VST3/AU).
- Rescan plugins in your DAW's preferences.
- Check plugin format compatibility for your system (Intel vs Apple Silicon).

## CPU Usage High

- LLAMAJA GC2020B is optimized for low CPU usage
- If you are using High Oversampling, try reducing it (e.g., from 8x to 2x or Off) during tracking.
- Increase your DAW's audio buffer size (e.g., 256 or 512 samples).

## Support Contact

For technical support or bug reports:

- **GitHub Issues**: github.com/DavidSignals/LlamajaGC2020B/issues
- **Email**: david.rabago.felix@gmail.com
- **Documentation**: Full documentation available on GitHub

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**David Rábago Félix**

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**Background**

Audio plugin developer and mechatronic engineer with a passion for creating tools that enhance creative workflows. Specializing in JUCE framework development and digital signal processing. My hands-on hardware expertise includes designing guitar pedal effects and building synthesizer circuits with integrated circuits and programmable platforms like Raspberry Pi, ESP32, Arduino, Teensy 4.1, and Daisy Seed. Currently designing a multi-effects pedalboard. Guitarist and music lover.

**Philosophy**

"Great tools should be invisible, they should empower creativity without getting in the way. Mono/Stereo Ears embodies this principle through its intuitive design and reliable performance."

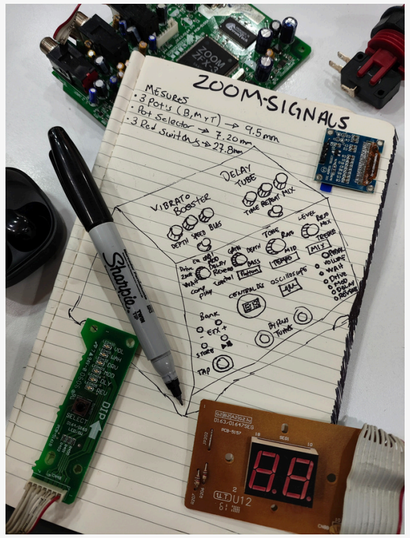
**Connect**

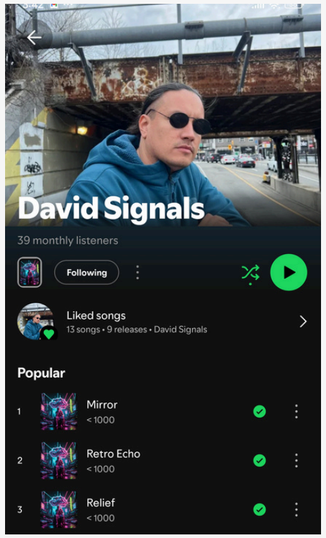
- GitHub: https://github.com/DavidSignals
- LinkedIn: www.linkedin.com/in/davidrabagofelix

**Acknowledgments**

Special thanks to the JUCE framework team and the audio developer community for their continuous inspiration and support.

# ABOUT ME



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# 📞 *SUPPORT & RESOURCES*

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